Ciphertext-Policy Attribute-based Encryption with Full Policy-Hiding for VANET's Data Engineering

Yanwei Zhou, Kui Ma, Guoji Song, Guangjin Zhang, Zirui Qiao, Xianxiang Liu, Yong Yu *Fellow, IEEE*

***Abstract*—Vehicular Ad-Hoc Networks (VANETs) demand secure and efficient data sharing, yet conventional Attribute-Based Encryption (ABE) schemes suffer from three critical limitations in this context: exposed access policies, cumbersome key revocation, and high computational overhead for vehicular users. To address these challenges, this paper proposes a novel Ciphertext-Policy ABE (CP-ABE) scheme tailored for VANETs, with three core contributions: (1) Full Policy-Hiding via a Linear Secret Sharing Scheme (LSSS) and an optimized hash chain, protecting both attribute values and policy structure; (2) Dynamic Key Revocation that operates seamlessly with hidden policies, enabling immediate invalidation of compromised credentials; (3) Efficient Outsourced Decryption that delegates intensive computation to a semi-trusted cloud, minimizing user overhead while preventing attribute exposure. The scheme is formally proven secure under standard cryptographic assumptions, achieving indistinguishability under chosen-plaintext attacks and full access structure anonymity. Comprehensive simulations validate its practical efficiency and scalability, demonstrating its suitability for real-world VANET deployments. This work advances the state of privacy-preserving data engineering by integrating policy confidentiality, immediate revocation, and lightweight decryption into a unified ABE framework for dynamic vehicular networks.**

***Index Terms*—ciphertext-policy attribute-based encryption, attribute revocation, fully hidden access policy, linear secret sharing scheme, outsourced decryption.**

I. Introduction

THE rapid evolution of intelligent transportation systems and connected vehicle technologies has positioned Vehicular Ad Hoc Networks (VANET) as a pivotal component in modern transportation infrastructure, facilitating both Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communications [1], [9], [37]. However, the inherent openness of wireless communication in VANETs introduces significant security vulnerabilities, including eavesdropping, spoofing, and privacy violations. Protecting sensitive vehicular data—such as real-time location, velocity, and driver identification is paramount to prevent malicious exploitation.

Manuscript received December 1, 2012; revised August 26, 2015. This work is supported by the National Natural Science Foundation of China (62502380, 62272287, U24B20149). Corresponding author: G. Song (Email: gjsong@hbun.edu.cn), Y. Yu (Email:yuyong@snnu.edu.cn).

Y. Zhou, G. Zhang, X. Liu and Y. Yu are with the School of Computer Science, Shaanxi Normal University, Xi'an, China.

K. Ma is with the School of Cyberspace Security, Xidian University, Xi'an, China.

G. Song is with the Information Center, Hebei Normal University for Nationalities, Chengde, China

Z. Qiao is with the School of Cyberspace Security, Xi'an University of Posts and Telecommunications, Xi'an, China.

Manuscript received April 19, 2021; revised August 16, 2021.

Traditional encryption methods, while effective for ensuring data confidentiality, lack the fine-grained access control capabilities required in dynamic VANET environments. Attribute-Based Encryption (ABE), particularly Ciphertext-Policy ABE (CP-ABE), addresses this limitation by enabling data owners to define expressive access policies based on attributes (e.g., Police Department, Emergency Vehicle) [6], [22], [24]. This ensures that only entities possessing the requisite attributes (e.g., authorized traffic authorities or emergency responders) can decrypt critical messages. The ABE scheme, first introduced by [5], extends traditional public key encryption by enabling fine-grained access control. ABE schemes are broadly categorized into two types: (1) Key-Policy ABE (KP-ABE). Access policies are embedded in user keys; (2) Ciphertext-Policy ABE (CP-ABE). Access policies are associated with ciphertexts.

Although ABE provides granular access control, exposed access policies can inadvertently leak sensitive information. For example, a policy such as Traffic Control Center, High-Priority may reveal that the sender is an emergency vehicle, making it susceptible to targeted attacks. Access Policy Hiding (APH) techniques mitigate this risk by obfuscating both the policy structure and its constituent attributes [15], [17], [39]. APH not only prevents inference attacks but also enhances user anonymity by disassociating specific attributes (e.g., Ambulance or Law Enforcement) from individual vehicles, thereby reducing the likelihood of tracking or impersonation.

VANET demands robust attribute revocation mechanisms to maintain security in dynamic scenarios. For instance, compromised credentials must be promptly revoked to prevent unauthorized access [12], [18], [38]. Furthermore, given the resource-constrained nature of vehicular nodes, outsourced decryption, where computationally intensive operations are delegated to edge or cloud servers, can significantly improve efficiency [8], [11], [13]. However, since cloud servers are semi-trusted, the decryption process must safeguard user attributes and secret keys, ensuring that partial decryption occurs without exposing sensitive information.

Existing ABE schemes fall short in addressing the unique challenges of VANET, which require a harmonious integration of policy hiding, dynamic revocation, and secure outsourced decryption. Our proposed CP-ABE scheme advances the state-of-the-art by concealing access policies to thwart adversarial exploitation, enabling efficient revocation to maintain system integrity, and supporting privacy-preserving outsourced decryption to optimize performance. This holistic approach ensures that VANET deployments achieve both stringent security and operational efficiency, paving the way for safer and more

reliable intelligent transportation systems.

A. Related Works

The CP-ABE scheme has emerged as the predominant approach in practical applications due to its natural alignment with real-world access control requirements. Recent studies [2], [7], [13], [16], [32] have significantly advanced ABE schemes, yet a critical limitation persists: current implementations largely neglect policy hiding mechanisms, potentially exposing sensitive user attributes and compromising privacy. The following text systematically analyzes recent developments and limitations in three key aspects of ABE schemes: policy hiding techniques, attribute revocation mechanisms, and outsourced decryption approaches. A comprehensive comparison of these features is presented in Table I, highlighting existing shortcomings and research opportunities.

TABLE I
REVIEW KEY TECHNOLOGIES AND DISADVANTAGES OF EXISTING PROPOSALS

Scheme	Key technology	Disadvantage
Scheme [2]	No hidden policy	Complete exposure of user privacy
Scheme [3]	Use time to revoke	Both the key and the ciphertext are changed
Scheme [11]	Outsourcing decryption	Exposing user attributes
Scheme [13]	No hidden policy	Complete exposure of user privacy
Scheme [14]	Partial hidden policy	Partial exposure of user privacy
Scheme [16]	No hidden policy	Complete exposure of user privacy
Scheme [19]	Full hidden policy	Insecure
Scheme [20]	Use time to revoke	Both the key and the ciphertext are changed
Scheme [26]	Outsourcing decryption	Exposing user attributes
Scheme [27]	Outsourcing decryption	Exposing user attributes
Scheme [29]	Outsourcing decryption	Exposing user attributes
Scheme [31]	Full hidden policy	Insecure
Scheme [33]	Partial hidden policy	Partial exposure of user privacy
Scheme [34]	Only revoke ciphertext	Unable to integrate policy hide
Scheme [39]	Partial hidden policy	Partial exposure of user privacy
Scheme [42]	Only revoke ciphertext	Unable to integrate policy hide

1) Access Policy Hiding: To enhance user privacy, several studies [14], [33], [39] have proposed ABE schemes with policy hiding capabilities, which can be implemented through two primary approaches: Masking the attributes comprising the policy (e.g., [10]); Hiding the policy structure itself (e.g., [35]). While partial policy hiding can protect attribute information, exposing policy structures may still compromise security. Although some works [21], [41] claim to achieve full policy hiding, their implementations only conceal attributes without obscuring the access structure. Ba *et al.* [4] proposed a blockchain-based solution, but it lacks a concrete smart contract implementation. Hao *et al.* [19] and Wu *et al.* [31] employed attribute Bloom Filters for complete policy hiding, yet [40] demonstrated security vulnerabilities in these approaches. This analysis reveals a critical research gap: most existing schemes fail to achieve genuine structural hiding while maintaining functional accessibility.

Challenge 1. *Designing an algorithm that simultaneously preserves access structure confidentiality and permits authorized decryptors to utilize the structure for data access remains a meaningful problem worthy of further investigation.*

2) Attribute Revoke: To enhance practical applicability, researchers have developed attribute revocation mechanisms to verify the legitimacy of attributes. Existing approaches typically employ timestamp-based revocation, where attribute keys are bound to timestamps during generation [3], [20], and revocation is performed by validating the authenticity of the timestamp. However, such methods require modifications to both user keys and ciphertexts, significantly increasing computational overhead. Recent advancements [16], [34] have proposed more efficient revocation schemes that enable attribute revocation through ciphertext updates and replace access policies without key modifications. Despite these improvements, a critical limitation remains: none of the existing revocation schemes adequately addresses the requirement for full policy hiding. The fundamental challenge arises because revocation typically operates through ciphertext modifications, and policy hiding obscures the very information needed for revocation. This creates an important research gap at the intersection of security and functionality.

Challenge 2. *Developing an efficient revocation algorithm that maintains full policy hiding, while preserving the system's revocation capabilities, represents a meaningful challenge worthy of investigation.*

3) Outsourcing Decryption: In practical deployment scenarios, users typically lack sufficient computational power and storage capacity to maintain encrypted data locally. Consequently, encrypted data is predominantly stored and processed in cloud environments. To address this paradigm, outsourced decryption techniques [13], [42] have been developed to facilitate secure and efficient access control over encrypted data. Notable contributions in this domain include the work of Liu *et al.* [27], who proposed an ABE scheme incorporating outsourced computation. Their approach delegates two critical decryption phases to trusted cloud servers: Verification of user attributes against access policies and recovery of split secret keys. This architecture significantly reduces the computational burden on end-users. Subsequent studies [11], [26], [29] have further advanced outsourced decryption capabilities. However, these existing solutions share a critical limitation: they necessitate the disclosure of user attributes to semi-trusted cloud servers during the decryption process. This requirement presents substantial privacy concerns, particularly when considering the fundamental assumption of cloud servers as semi-trusted entities, the inherent conflict with policy hiding objectives, and the potential for attribute leakage, compromising user privacy.

Challenge 3. *Developing an outsourced decryption method that preserves the efficiency benefits of cloud-based computation while completely eliminating the exposure of user attributes to service providers remains a question demanding further investigation.*

B. Our Motivations and Contributions

For a CP-ABE scheme to be effectively deployed in VANET, it must incorporate three critical features: full access policy hiding, efficient attribute revocation, and secure outsourced decryption. However, our analysis reveals that most existing constructions fail to adequately address the following key challenges: (1) Current schemes have not successfully achieved the simultaneous implementation of full access policy hiding and attribute revocation. (2) Even when supporting

attribute revocation and policy hiding, most solutions either lack outsourced decryption capabilities or compromise privacy by exposing user attributes during the decryption process. (3) No existing scheme concurrently satisfies all three requirements: attribute revocation, policy hiding, and outsourced decryption, while maintaining computational efficiency and provable security.

To overcome these limitations, we propose a novel CP-ABE scheme that seamlessly integrates full policy hiding, dynamic attribute revocation, and privacy-preserving outsourced decryption. Our solution offers enhanced security for vehicle data access control and reduces the computational burden on resource-constrained devices in VANET. Building upon the identified gaps in prior art, particularly the limitations of schemes like [33] and [40], our work makes the following fundamental contributions, which are the first to cohesively achieve all three objectives:

(1) *Full Policy Hiding via an Obfuscated LSSS Structure, overcoming partial hiding*. Unlike the *partial* policy hiding (which only conceals attributes) achieved in schemes like [14], [33], [40], we implement *complete* policy hiding. This is realized through a novel dual-layer method: (i) attribute value concealment using cryptographic hash functions, and (ii) critical to our contribution, the obfuscation of the Linear Secret Sharing Scheme (LSSS) access matrix structure via an optimized hash chain table. This approach ensures that both the attribute set and the policy's logical structure remain entirely hidden, addressing a core limitation of previous partial-hiding solutions.

(2) *A Revocation Mechanism Intrinsically Compatible with Fully Hidden Policies*. A key challenge unaddressed by prior schemes, including [33] and [40], is executing efficient revocation when the policy is completely hidden. We introduce a dynamic revocation mechanism that operates directly on the obfuscated ciphertext structure. The Trusted Authority (TA) can revoke attributes by filtering ciphertext components based on a secure list, without needing to recover or expose the original policy. This ensures immediate invalidation of compromised credentials while strictly maintaining full policy confidentiality, resolving the hiding-revocation conflict inherent in previous works.

(3) *Privacy-Preserving Outsourced Decryption Guaranteed under Hidden Policies*. Existing outsourced decryption schemes (e.g., [11], [26]) often require the cloud server to know user attributes, creating a privacy leak. Our scheme overcomes this by designing a secure two-phase decryption protocol that functions under fully hidden policies. The cloud server performs all heavy computations without learning any information about the user's attributes or the access policy. The user then recovers the plaintext with minimal effort. This design closes the attribute exposure vulnerability, providing a stronger security guarantee tailored for VANETs.

The technical soundness and practical efficacy of our integrated approach are rigorously demonstrated through detailed cryptographic constructions, comprehensive security proofs, and performance evaluations. More importantly, its fundamental novelty lies in the synergistic co-design of the above components, which solves the intrinsic conflicts between the three features. This makes it particularly suitable for VANET applications where privacy, dynamism, and performance are critical requirements.

II. PRELIMINARIES

We use Table II to describe the definition of symbols used in this paper.

TABLE II
COMMON SYMBOLS

Symbol	Definition
$\mathbb{A}$	access structure
$\boldsymbol{M}_{n\times l}$	matrix with n rows and l columns
H_1, H_2, H_3	secure one-way hash functions
κ	security parameter
ID	vehicle identification
$params$	system public parameter
S	attribute set
S_a	attribute space
msk	master secret key
PAK	attribute key
UIM_d	identity mark
c	attribute
CT	origin ciphertext
CT_s	ciphertext
CT_f	revoked ciphertext
ODK	outsourcing decryption key
E	outsourced decryption of ciphertext
m	vehicle data

A. Access Structures

Let $S_a = \left\{S_{a_1}, \cdots, S_{a_n}\right\}$ be an attribute universe and has a monotone collection $\mathbb{A} \in 2^{\{S_{a_1},\cdots,S_{a_n}\}}$, for any B, C, if $B \in \mathbb{A}$ and $B \subseteq C$, then $C \subseteq \mathbb{A}$, so $\mathbb{A}$ is called a collection of non-empty subsets of $S_a = \left\{S_{a_1}, \cdots, S_{a_n}\right\}$. Then, $\mathbb{A}$ is an access structure, and the set in $\mathbb{A}$ is called the authorized set; otherwise, it is called an unauthorized set.

B. Linear Secret Sharing Scheme

LSSS is employed to turn an access structure $\mathbb{A}$ on S_a into a matrix $\boldsymbol{M}$ with size $n \times l$. We use a vector $\vec{\boldsymbol{V}} = \left(\eta, y_2, \cdots, y_l\right)^\top$ to hide a secret value $\eta \in Z_q^*$, where $y_2, y_3, \cdots, y_l \in_R Z_q^*$. Let $\boldsymbol{M}_{n\times l} \times \vec{\boldsymbol{V}} = \vec{\boldsymbol{N}}$. There exists a function $\rho(i) : \{1, \cdots, n\} \to N$ that maps a row to an attribute, and it has the following properties:

Definition $L = \left\{i : \rho(i) \in S\right\}$ and $|L| < n$. For any authorized set $\mathbb{A}$, there exists a vector $\vec{\boldsymbol{\omega}} = \left\{\omega_i \in Z_q^* | i \in L\right\}$ satisfying $\boldsymbol{M}_{n\times l}^\top \vec{\boldsymbol{\omega}} = \left(1, 0, \cdots, 0\right)$, so it have $\Sigma_{i\in L} \boldsymbol{M}_i^\top \omega_i = \eta$ (where the value of ω_i is not unique). Otherwise, for any unauthorized set, there not exists a vector $\vec{\boldsymbol{\omega}} = \left\{\omega_i \in Z_q^* | i \in L\right\}$ satisfying $\boldsymbol{M}_{n\times l}^\top \vec{\boldsymbol{\omega}} = (1, 0, \cdots, 0)$.

C. Complexity Assumption

Now, we define decisional p-Parallel Bilinear Diffie-Hellman Exponent (p-$PBDHE$) assumption in the following. Let G be a multiplicative cyclic group, q is the order of G, and g is a generator of G, the bilinear pairing $e : G \times G \rightarrow G_T$. For $\alpha, s, \beta_1, \beta_2, \cdots, \beta_p \in Z_q$ and any probabilistic polynomial time (PPT) adversary $\mathcal{A}$, it is difficult to distinguish $e(g,g)^{\alpha^{p+1}s}$ from a random element $R \in G_T$ given the following parameters.

$$\mathbb{D} = \Big\{ g, g^s, g^\alpha, g^{\alpha^2}, \cdots, g^{\alpha^p}, g^{\alpha^{p+2}}, \cdots, g^{\alpha^{2p}}, \Big\{ g^{s\beta_1}, g^{\alpha/\beta_1}, g^{\alpha^2/\beta_2}, \cdots, g^{\alpha^p/\beta_p}, g^{\alpha^{p+2}/\beta_2}, \cdots, g^{\alpha^{2p}/\beta_p} \Big\}, \Big\{ g^{\alpha s \beta_k/\beta_j}, \cdots, g^{\alpha^i s \beta_k/\beta_j} \Big\}_{1 \le i,j,k \le p, k \ne j} \Big\}$$

The advantage $\mathsf{Adv}_{\mathcal{A}}(\kappa)$ of the adversary $\mathcal{A}$ in solving p-$PBDHE$ problem can be defined as

$$\mathsf{Adv}_{\mathcal{A}}(\kappa) = \left| \Pr\left[\mathcal{A}\Big(D, e(g,g)^{\alpha^{p+1}s}\Big)\right] - \Pr\left[\mathcal{A}\Big(D, R\Big)\right] \right|$$

Then, we have that $\mathsf{Adv}_{\mathcal{A}}(\kappa) \le \mathsf{negl}(\kappa)$, where $\mathsf{negl}(\kappa)$ is a negligible value over κ.

D. Fully Hidden Access Structure

In LSSS-based ABE schemes, achieving comprehensive policy hiding requires protection of two critical components: (1) the attribute values specified in the access policy, and (2) the transformed policy matrix structure. Our proposed solution addresses these requirements through dual cryptographic mechanisms:

(1) Attribute Value Concealment. We employ a cryptographically secure one-way hash function to obfuscate policy attributes, establishing an irreversible mapping between original attributes and their hashed representations. This transformation preserves the semantic meaning of attributes while preventing their disclosure during policy evaluation.

(2) Access Structure Obfuscation. The fundamental challenge in hiding access structures stems from the necessity of using both the access matrix $\boldsymbol{M}_{n\times l}$ and mapping function ρ during decryption. Our novel solution introduces a hierarchical hash chain table (HT) maintained by the authority, as illustrated in Fig. 1. The system utilizes a cryptographic hash function H_2, ensuring one-way transformation, the mapping $\rho(i)$ associates matrix rows with attributes, unique encryptor identifiers (ID) prevent policy collisions, and timestamps (t) for temporal differentiation of policies.

The hash chain table stores complete matrix row information ($i||\boldsymbol{M}_{i,:}$) while enabling authorized decryptors to securely retrieve necessary components by computing $H_2(\rho_i, ID, t)$. For policies containing duplicate attributes, the sub-list structure maintains proper ordering through sequential storage of encryptor with identity ID. During decryption, the system retrieves all relevant $i||\boldsymbol{M}_{i,:}$ pairs from corresponding sub-lists when present, ensuring correct policy evaluation while maintaining full structure confidentiality.

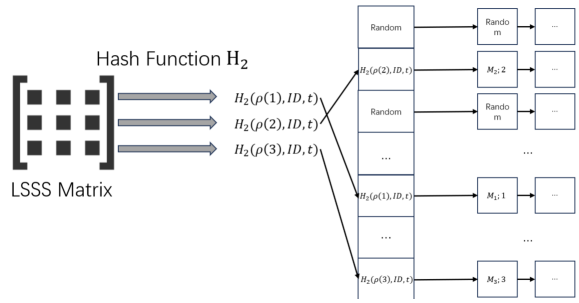

Fig. 1. Fully hidden access policy structure diagram

III. Framework and Security Model of VANET

This section presents the system framework of the VANET and elaborates on its associated security requirements.

A. System Framework

The VANET architecture comprises five fundamental components, as illustrated in Fig. 2: On-Board Units (OBU) installed in vehicles, Roadside Units (RSU) deployed along transportation infrastructure, a TA serving as the central security manager, a Vehicular Cloud (VC) providing distributed computing resources, and Service Providers (SP) delivering various vehicular applications. These entities collectively form a hierarchical network structure that enables secure and efficient vehicle communications while supporting diverse intelligent transportation services.

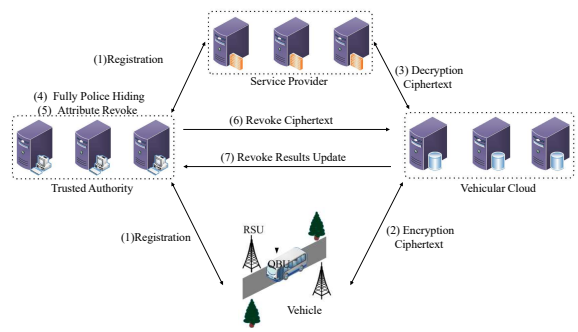

Fig. 2. System Framework and Data Engineering of VANET

(1) ***TA***. The TA serves as the central security and authentication entity in the system, responsible for issuing digital certificates to OBU and RSU to mitigate malicious activities (e.g., spoofing, fake messages), generating public parameters for encryption, and a master secret key for cryptographic operations. Furthermore, producing attribute-specific private keys for users based on their attribute sets, and maintaining an attribute information table TB to track and validate user attributes.

(2) ***VC***. Modern VANETs incorporate cloud computing infrastructure to facilitate distributed data storage and computationally intensive processing tasks. This integration supports critical real-time applications, including traffic flow prediction, dynamic route optimization, and multimedia service delivery. Within this architecture, the VC operates as an honest-but-curious entity that maintains both the original ciphertext

and updated ciphertext while faithfully executing designated cryptographic algorithms. Within this architecture, the VC is primarily modeled as an *honest-but-curious* entity: it faithfully executes the designated cryptographic algorithms but may attempt to learn information from the data it processes. Furthermore, to assess robustness, we also consider a stronger adversarial model where the VC could be *malicious* during the outsourced decryption phase, potentially returning incorrect computation results to disrupt the service or learn from the user's reaction.

(3) ***OBU***. As an embedded device installed in vehicles, the OBU incorporates dedicated short-range communications technology to enable reliable V2V and V2I interactions. This unit performs real-time processing of sensor-derived parameters while simultaneously managing secure message exchange with proximate vehicles and roadside infrastructure. Within the proposed cryptographic framework, vehicles function as encrypting entities that generate ciphertexts by applying designated access policies and the system's master public key during the encryption process.

(4) ***RSU***. RSUs are stationary infrastructure components deployed at key transportation locations, including road segments, intersections, and toll collection points. These units serve as critical communication gateways while providing internet access and integration with centralized traffic control systems. Furthermore, RSUs significantly improve network coverage reliability and perform essential emergency message dissemination functions, such as broadcasting critical safety alerts regarding traffic accidents or hazardous road conditions.

(5) ***SP***. SPs function as trusted entities that deliver vehicular services, including road condition forecasting. As authorized decryptors, SP first obtain their attribute-based private key from the TA and retrieve the partially decrypted ciphertext from the VC.

B. Security Definition

Our proposed scheme must satisfy several fundamental security requirements, including data confidentiality, efficient attribute revocation, user anonymity, complete policy hiding, and collusion resistance. These security properties collectively ensure the robustness of our cryptographic framework. Data confidentiality and anonymity guarantee protection against unauthorized information disclosure, while collusion resistance prevents coordinated attacks from multiple malicious parties. The attribute revocation mechanism effectively prevents compromised or expired attributes from being utilized in decryption operations. Furthermore, our policy hiding implementation ensures comprehensive protection of both attribute sets and access policy structures throughout all cryptographic operations, making them inaccessible to potential adversaries during encryption and decryption processes.

To meet the above security objectives, we have described the main security definitions:

(1) ***Data Confidentiality***. Data confidentiality represents a fundamental security property that safeguards sensitive information from unauthorized access, utilization, or dissemination. As a cornerstone of information security principles, it assumes particular significance for protecting private vehicular data in VANET environments. This critical security guarantee ensures that no PPT adversary can successfully compromise, exploit, or reveal protected data through cryptographic analysis or system vulnerabilities.

(2) ***Anonymity***. Anonymity refers to the property whereby the identity and distinguishing characteristics of participating entities remain undisclosed within a system. In cryptographic systems, anonymity guarantees that no PPT adversary can derive or infer user-specific information through analysis of publicly accessible data, including ciphertexts, access policies, or transaction records.

(3) ***Collusion Resistant***. In the ABE scheme, collusion resistance prevents any coalition of users from combining their partial attribute keys to construct a valid decryption key. Specifically, the cryptographic design ensures that no PPT adversary can successfully merge or manipulate distributed attribute keys from multiple users to forge a functional key that would enable unauthorized decryption.

(4) ***Backward and Forward Security***. Forward and backward security constitute essential cryptographic properties in the design of secure communication protocols. Forward security ensures the confidentiality of previously established communications remains protected even if long-term secret keys are compromised in the future. Conversely, backward security guarantees that future communication sessions maintain their integrity and confidentiality regardless of any security breaches that may have occurred in past sessions. These complementary security properties work in tandem to provide comprehensive temporal protection for communication systems, with forward security safeguarding historical data and backward security shielding subsequent interactions from the effects of prior compromises.

IV. CP-ABE Scheme

A. Formal Definition

In this section, we provide the formal definition of a CP-ABE scheme with full policy-hiding and outsourcing decryption.

(1) $(Params, msk) \leftarrow \mathsf{Setup}(1^\kappa)$. TA inputs a security parameter κ and generates the public parameter $Params$ and the master secret key msk. Specifically, the list of user attributes TB is a system parameter and included in $Params$. In addition, $Params$ is a common input to the following algorithms, and we will omit $Params$ in the description of the following algorithms for convenience.

(2) $PAK \leftarrow \mathsf{KeyGen}(S, msk)$. The user sends its own attribute set S to the TA, and the TA generates an attribute private key PAK for the user by using the master secret key msk, and sends it to the user to complete the registration.

(3) $\tilde{c} \leftarrow \mathsf{Attribute\text{-}Hiding}(c, Fun)$. TA hides attribute c by means of an attribute hiding function Fun, and output the hidden attribute $\tilde{c}$. In particular, in our proposal, we use a one-way collision-resistant hash function $H_3 : \{0,1\}^* \rightarrow Z_q^*$ to implement attribute hiding, which ensures better computational efficiency of our scheme.

(4) $CT_s \leftarrow \mathsf{Enc}(m, \mathbb{A})$. The user employs the access policy $\mathbb{A}$ to generate the ciphertext on the message m. Furthermore, to

complete the hiding of the policy, the user encrypts ciphertext with the public key and finally generates the ciphertext CT_s.

(5) $CT_f \leftarrow \mathsf{Attribute\text{-}Revoke}(TB, UIM_d, CT_s, msk)$. The cloud server VC sends the decryptor identity UIM_d and the ciphertext CT_s to the TA, which decrypts the ciphertext using msk, runs the revocation algorithm in conjunction with the user attribute list TB, and outputs the revoked ciphertext CT_f.

(6) $CT'_f \leftarrow \mathsf{Policy\text{-}Structure\text{-}Hiding}(HT, \boldsymbol{M}_{n_f \times l}, \rho_f, ID)$. TA obtains the encrypted access policy $\boldsymbol{M}_{n_f \times l}$, mapping function ρ_f, and user identity ID, afterwards, hides the access policy using our hash chaining table HT, and outputs the final ciphertext CT'_f.

(7) $m \leftarrow \mathsf{Dec}(CT'_f, ID, ODK, c, PAK)$. The decryptor inputs the encryptor's identity ID, its own attribute c, decryption key PAK, and outsourced decryption key ODK, and outputs the plaintext m using the VC outsourced decryption technique.

To facilitate the description of the security model, we provide a specific formal definition of each algorithm of the CP-ABE scheme. However, in the concrete construction of our CP-ABE scheme, we perform an implicit investigation of the above algorithms, taking the actual requirement of VANET as an example, and the specific operation process is described in detail.

B. Security Model

Our CP-ABE scheme needs to hold a security, namely selective CPA security, and we will define it in detail below. This game is played between a challenger $\mathcal{C}$ and a PPT adversary $\mathcal{A}$, which proves that our system has selective CPA security. Before the game starts, the adversary $\mathcal{A}$ sends the access structure $\mathbb{A}^*$ to the challenger $\mathcal{C}$.

Setup. The challenger $\mathcal{C}$ runs the initialization algorithm $(Params, msk) \leftarrow \mathsf{Setup}(1^\kappa)$ and sends the system's public parameters $Params$ to the adversary $\mathcal{A}$. Additionally, $\mathcal{C}$ initializes an internal revocation state $\mathcal{R}$ (simulating the TA's attribute table TB) to consistently track revoked attributes for any user during the game.

Query Phase 1. The adversary $\mathcal{A}$ may adaptively issue two types of queries to the challenger $\mathcal{C}$:

(1) *Private Key Query*. $\mathcal{A}$ submits an attribute set S, where S does not satisfy the challenge access structure $\mathbb{A}^*$. $\mathcal{C}$ runs $PAK \leftarrow \mathsf{KeyGen}(S, msk)$ and returns the attribute key PAK to $\mathcal{A}$.

(2) *Revocation Query*. $\mathcal{A}$ submits a message pair (c, UIM_d), where c is an attribute and UIM_d is a user identifier. $\mathcal{C}$ updates its internal revocation state $\mathcal{R}$ for UIM_d with respect to c (i.e., marks c as revoked for UIM_d). This simulates the real-world Attribute-Revoke algorithm. $\mathcal{C}$ returns an acknowledgment to $\mathcal{A}$.

Challenge. $\mathcal{A}$ selects two equal-length messages m_0, m_1 and sends them to $\mathcal{C}$. Then, $\mathcal{C}$ selects $b \in \{0, 1\}$ and encrypts the message m_b with the access structure $\mathbb{A}^*$ to generate the ciphertext CT_b, where $CT_b \leftarrow \mathsf{Enc}(m_b, \mathbb{A}^*)$. Finally, $\mathcal{C}$ sends the ciphertext CT_b to $\mathcal{A}$. The generation of CT_b is consistent with the current revocation state $\mathcal{R}$, modeling a ciphertext that exists prior to any user-specific revocation applied via subsequent queries.

Query Phase 2. $\mathcal{A}$ may continue to adaptively issue ***Private Key Queries*** and ***Revocation Queries*** as in Phase 1, under the same restriction that the queried attribute sets S for key generation do not satisfy $\mathbb{A}^*$.

Guess. $\mathcal{A}$ outputs a guess b'. If $b' = b$, then the adversary $\mathcal{A}$ wins this game; otherwise, $\mathcal{A}$ fails.

We define the advantage of $\mathcal{A}$ in the selective CPA security game as $\mathsf{Adv}_{\mathcal{A}}(\kappa) = \left| \Pr[b' = b] - \frac{1}{2} \right|$.

Definition 1 (Selective CPA Security)**.** *If the advantage* $\mathsf{Adv}_{\mathcal{A}}(\kappa)$ *is negligible, then we say that the corresponding CP-ABE scheme has selective structure CPA security.*

Specially, in our security model, TA is modeled as fully trusted. It correctly executes the Setupand KeyGenalgorithms. The adversary in the security game is not allowed to corrupt the TA or obtain the master secret key mskdirectly, which reflects the real-world assumption that the TA is a hardened, trusted entity. The TA's ability to process ciphertexts (for hiding and revocation) is notmodeled as a decryption oracle available to the adversary, because this processing occurs outside the interactive challenge phase and does not provide the adversary with adaptive decryption queries on arbitrary ciphertexts.

V. Our Concrete Construction of CP-ABE Scheme

This section provides the detailed construction of our CP-ABE scheme.

A. Initialization

TA selects a security parameter κ to initialise the system. The declared parameters are employed by all algorithms in the system. AC selects two cyclic groups G, G_T of prime order q, with a bilinear pairing $e : G \times G \rightarrow G_T$, g is a generator of G. TA chooses three secure one-way hash functions $H_1 : \{0,1\}^* \rightarrow G$, $H_2 : \{0,1\}^* \rightarrow Z_q^*$ and $H_3 : \{0,1\}^* \rightarrow Z_q^*$. Note that whenever the input to a hash function is not a bit string (e.g., a field element from Z_q^* or an integer), it is first converted into a deterministic string representation (e.g., its binary) before being hashed. Furthermore, TA chooses $\alpha, \beta \in Z_q^*$ and sets master secret key msk and system public parameters $params$ as follow.

$$Params = \{G, G_T, g, e, g^\alpha, e(g,g)^\beta, \mathsf{SE}\} \text{ and } msk = \{\alpha, \beta\}$$

where $\mathsf{SE} = \big(\mathsf{SE.Enc}(), \mathsf{SE.Dec}()\big)$ is a semantically secure asymmetric encryption and decryption algorithms.

TA creates an empty user attribute information table T_B, as shown in Table III, where each row of the table corresponds to the attribute situation corresponding to each user. The first column of the table represents the real identity of the user, and the subsequent columns represent the attributes. Furthermore, Y denotes that the user has the corresponding attribute, and N denotes that the user does not have the corresponding attribute.

B. Registration and Attribute Hiding

When a SP wants to join the scheme, they should register with TA. Firstly, the SP sends attribute set S to TA, after

TABLE III
THE INITIAL TABLE OF USER INFORMATION MAINTAINED BY TA

$Identity$	$H_3(c_1)$	$H_3(c_2)$	$\cdots$	$H_3(c_i)$
ID	Y	N	$\cdots$	Y

that, TA selects $t \in Z_q^*$ and a unique identity mark $UIM_d \in \{0,1\}^*$ and it compute the SP attribute key PAK as follow. TA hides the attribute value using a hash function, and we use $H_3(c)$ to denote the attribute, where c represents an attribute. In this case, the attribute of the policy is hidden.

AC choose $t \in Z_q^*$ to compute

$$K = g^{\beta} g^{\alpha t}, \quad K' = g^t \quad \text{and} \quad K_{c_i} = (H_1(H_3(c_i)))^t,$$

where $c_i \in S$. After that, sends $PAK = \{UIM_d, K, K', K_{c(c \in S)}\}$ to the SP.

If the SP has been registered, then TA changes the table T_B as shown in Table IV. If an attribute is revoked for a SP, then TA changes the attribute column for that SP in this table directly to N.

TABLE IV
THE TABLE OF USER INFORMATION MAINTAINED BY TA

$Identity$	$H_3(c_1)$	$H_3(c_2)$	$\cdots$	$H_3(c_i)$
UIM_d	Y	Y	$\cdots$	Y

C. Encryption

When the vehicle with identity ID needs treatment, the vehicle data m is encrypted using an access policy $\mathbb{A}$ that suits their needs. When the vehicle needs treatment, the vehicle data m is encrypted using access police $\mathbb{A}$ that suits their needs. We partition the access police $\mathbb{A}$ via the LSSS as $(\boldsymbol{M}_{n\times l}, \rho)$, where $\boldsymbol{M}_{n\times l}$ denotes the access policies described with matrix form, ρ is the correspondence between rows and attributes of the matrix after the secret partition. The vehicle randomly selects a vector $\overrightarrow{\boldsymbol{V}} = \left(\eta, y_2, y_3, \cdots, y_l\right)^{\top}$.

For i from 1 to n, computes $\delta_i = \boldsymbol{M}_{n\times l} \cdot \overrightarrow{\boldsymbol{V}}$. The vehicle randomly selects $(r_1, r_2, \cdots, r_n) \in Z_p^*$ to encrypt vehicle data m as follows.

$$\begin{aligned}
&C_i = g^{\alpha\delta_i} \cdot H_1(\rho(i))^{-r_i}, \quad \text{where} \quad i = 1, \cdots, n \\
&C_i' = g^{r_i}, \quad \text{where} \quad i = 1, \cdots, n \\
&C = g^{\eta} \\
&\widetilde{C} = m \cdot e(g,g)^{\eta\beta}
\end{aligned}$$

The vehicle outputs original ciphertext

$$CT = \left\{(C_i, C_i')_{i=1,\cdots,n}, C, \widetilde{C}, \boldsymbol{M}_{n\times l}, \rho, ID\right\}.$$

The vehicle uses TA's public key (public parameter) $params$ to encrypt CT by using the algorithm SE.Enc(), and only TA's private key (master secret key) msk can be correctly decrypted. Finally, the vehicle sends ciphertext CT_s to VC for storage.

D. Attribute Revoke and Structure Hiding

The processes described in this subsection are *post-encryption* enhancements performed by the TA. The input CT_s is the encapsulated original ciphertext that was independently generated and uploaded by the data owner (vehicle) during the encryption phase (Section V.C). The TA's role is to apply advanced security features—policy hiding and dynamic revocation—onto this *existing* ciphertext, without altering the access control policy originally defined by the data owner.

When an SP requests decryption, it sends UIM_d to VC. The cloud server VC then forwards a tuple (UIM_d, CT_s) to TA. Upon receiving it, the TA first unwraps CT_s using its master secret key to obtain the original ABE ciphertext CT. This step is necessary solely for the TA to access the internal structure of CT (i.e., components C_i, C_i', and matrix M) to perform policy hiding and revocation; the TA gains no information about the plaintext message m from this operation. Subsequently, the TA verifies the SP's attribute legitimacy against the user attribute information table TB and updates it. If the SP lacks a specific attribute, the corresponding entry in the table is set to N, as illustrated in Table V. The TA then executes the revocation algorithm detailed below (Algorithm 1).

TABLE V
USER INFORMATION UPDATE TABLE MAINTAINED BY TA

U_{id}	$H_3(c_1)$	$H_3(c_2)$	$\cdots$	$H_3(c_i)$
ID	N	N	$\cdots$	Y

Algorithm 1 The algorithm of Attribute Revocation

Input: CT_s: The ciphertext to be decrypted
$TB[UIM_d]$: The corresponding row data with the identity UIM_d
Output: CT_f: Updated ciphertext
//i means the row index of access structure $\boldsymbol{M}_{n\times l}$.
//$\rho(i)$ means the hashing of attribute values corresponding to row i of the access matrix $\boldsymbol{M}_{n\times l}$.
for i from 1 to n **do**
  if $TB[U_{id}][\rho(i)] = N$ **then**
    1. Remove the row of index i in $\boldsymbol{M}_{n\times l}$.
    2. Remove the items $\{C_i, C_i'\}$ in CT_f.
  end if
end for
// We define the update access structure as $(\boldsymbol{M}_{n_f\times l}, \rho_f)$

Following attribute revocation, the TA implements policy hiding using an improved hash chain. The TA constructs a hash table HT of length $2n_f$ (assuming the hash function's range is $2n_f$). For each $i = 1, \cdots, n_f$, the TA computes $HIN_i = H_2(\rho_f(i), ID, t)$. The entry at index HIN_i in HT stores the row vector of $\boldsymbol{M}_{n_f\times l}$ corresponding to $\rho_f(i)$. Empty slots in HT are filled with random values. For distinct attributes, the mapping addresses differ, and data is stored accordingly. To handle policies with duplicate attributes, a sub-list structure is employed within HT (as introduced in Section II.D and

shown in Fig. 3). When attributes, identity, and timestamp are identical, the corresponding information is stored sequentially in the sub-list.

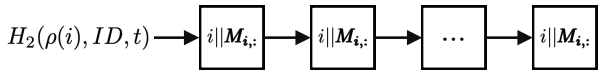

Fig. 3. The sub-list framework of HT

These operations ensure the access structure remains hidden: entities other than the TA cannot recover the matrix $\boldsymbol{M}_{n_f \times l}$. An authorized decryptor can only retrieve the necessary data based on its own attributes, thus keeping the policy structure fully confidential. At this stage, the TA has completed the full policy hiding process.

Finally, the TA sends the updated and policy-hidden ciphertext $CT_f = \{(C_i, C_i')_{i=1,\cdots,n_f}, C, \widetilde{C}, HT, ID, t\}$ to VC for storage and subsequent outsourced decryption.

E. Outsourcing Decryption

After downloading the ciphertext from VC, the SP uses the hash function H_2 to find the row vector of the access matrix $\boldsymbol{M}_{n_f \times l}$ and row index i from HT. When the HT subchain stores data, the SP fetches all the data in turn until the appropriate vector is found or the subchain query ends. The SP removes the first character of $i||\boldsymbol{M}_i$ to finds the vector $\overrightarrow{\boldsymbol{\omega}}$ matching the condition $\boldsymbol{M}^{\top}_{n_f^* \times l} \times \overrightarrow{\boldsymbol{\omega}} = (1, 0, \cdots, 0)^{\top}$, if vector $\overrightarrow{\boldsymbol{\omega}}$ exists, the SP's attribute conforms to the access policy and saves vector $\overrightarrow{\boldsymbol{\omega}}$. Otherwise, it does not. To reduce calculation costs, the SP sets the row index i corresponding to the matrix $\boldsymbol{M}_{n_f^* \times l}$ to I. In this process, the SP is unaware of the legitimacy of the row vector found based on its attributes, which ensures that the structure and attribute values of the access policy are not available to any entity other than TA.

To reduce the SP's computational consumption, the SP outsources the complexity of the computation to the cloud server by outsourcing the decryption algorithm without revealing their attributes and keys, and performing only simple calculations to complete the decryption themselves.

The SP defines collection $I = \{i\} \subset \{1, 2, \cdots, n_f\}$ and randomly selects outsourcing decryption key $ODK = z \in Z_q^*$ and calculates

$$K_o = K^z$$
$$K_o^{'} = K^{'z}$$
$$K_{o,c} = K_c^z (c \in S_f)$$

The SP sends $\{K_o, K_o', K_{o,c}(c \in S_f), \vec{\boldsymbol{\omega}}\}$ to VC.

VC calculates

$$E = \frac{e(C, K_o)}{\prod_{i \in I}(e(K_{o,\rho_f(i)}, C_i')e(K_o', C_i))^{\boldsymbol{\omega_i}}}$$

and outputs E to SP.

The SP obtains the decryption parameter E from the cloud server and creates the final plaintext message m with the outsourcing decryption key $ODK = z$ by performing the following calculation.

The SP calculates $m = \dfrac{\widetilde{C}}{E^{1/z}}$.

The protocol above assumes the cloud server is semi-honest (honest-but-curious). In a stronger adversarial model where the server may be malicious and return an arbitrary or incorrect value $\hat{E} \neq E$, the user's final decryption will output an invalid message or random bits. This provides abort-safety: a malicious server can cause a denial-of-service (decryption failure) but cannot, undetected, force the user to accept an incorrect plaintext message. The user will be aware that the process has failed, though it cannot distinguish between failure due to lack of attributes and failure due to a malicious server.

To achieve verifiability (i.e., allowing the user to detect and prove malicious behavior), standard techniques like requesting computation on randomized ciphertexts can be integrated into our framework. This would introduce additional rounds or computations for challenge generation and verification. In this work, prioritizing minimal overhead for the resource-constrained user, we present the core, efficient outsourcing protocol. The exploration of optimal verifiable extensions for this specific construction, balancing efficiency and security in the fully malicious model, is an important direction for future work.

VI. Soundness Analysis of Our CP-ABE Scheme

In this section, we will prove the correctness of our scheme and demonstrate the security properties defined in our security model, including unforgeability, anonymity, and traceability.

A. Correctness

The correctness of our proposal is obtained from the following equation.

$$\begin{aligned} E &= \frac{e(C, K_o)}{\prod_{i \in I}(e(K_{o,\rho_f(i)}, C_i')e(K_o', C_i))^{\boldsymbol{\omega}_i}} \\ &= \frac{e(g^{\eta}, g^{\beta} g^{\alpha t})^z}{\prod_{i \in I} e(g, g)^{\alpha t \delta_i \boldsymbol{\omega}_i z}} = \frac{e(g, g)^{\eta \beta z} e(g, g)^{\alpha \eta t z}}{e(g, g)^{\alpha t \eta}} = e(g, g)^{\eta \beta z} \end{aligned}$$

B. Security Proof

Theorem 1. *If the decisional p-BDHE assumption holds in the random oracle model, then for any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ participating in the selective CPA security game, the advantage $\mathsf{Adv}_{\mathcal{A}}(\kappa)$ in distinguishing the challenge ciphertext is negligible in the security parameter κ. In other words, under the stated assumption and model, the proposed scheme achieves selective CPA security.*

Proof. Suppose we have a PPT adversary $\mathcal{A}$ with non-negligible advantage $\mathsf{Adv}_{\mathcal{A}}(\kappa)$ in the selective security game break our construction and it choose a challenge matrix $\boldsymbol{M}^*$ where both rows and columns are at most q. We construct a decisional p-$BDHE$ problem game to be played between a challenger $\mathcal{C}$ and the adversary $\mathcal{A}$.

Initialization. The challenger $\mathcal{C}$ takes in a decisional p-$BDHE$ challenge $(\mathbb{D}, T)$, where $T = e(g, g)^{\alpha^{p+1} s}$ or $T \in G_T$. $\mathcal{A}$ outputs challenge access structure $(\boldsymbol{M}^*, \rho^*)$ to $\mathcal{C}$, where $\boldsymbol{M}^*$ has $n^* \leq q$ rows and $l^* \leq q$ columns.

Setup. $\mathcal{C}$ randomly selects $s' \in Z_q^*$ and implicitly sets $s = s' + \alpha^{q+1}$ by letting $e(g,g)^s = e(g^\alpha, g^{\alpha^q})e(g,g)^{s'}$.

We set up a random oracle H_1 and use table L_r to store the values corresponding to $H_1(x_i)$, where $(x_i = H_3(c_i))$. We set the set B to be the index of the corresponding row of the attribute in the challenge matrix $\boldsymbol{M}^*$, $B = \left\{ i \middle| \rho^*(i) = x \right\}_{i=1}^{n^*}$. When the value x asked for exists in L_r, the challenger $\mathcal{C}$ extracts $H_1(x)$ from the table and sends it to $\mathcal{A}$, if not, the challenger randomly seclcts η_x for x to constructs the random oracle H_1 according to the following rules:

$$H_1(x) = g^{\eta_x} \prod_{i \in B} g^{\alpha M^*_{i,1}/\beta_i} \cdot g^{\alpha^2 M^*_{i,2}/\beta_i} \cdots g^{\alpha^{l^*} M^*_{i,l^*}/\beta_i}$$

When the set B is the empty set, $H_1(x) = g^{\eta_x}$. We easily know that the output of H_1 in this case is also randomly distributed.

Query Phase 1. $\mathcal{A}$ sends an attribute S to $\mathcal{C}$ to generate attribute key, the attribute set S does not match challenge access structure $\boldsymbol{M}^*$. Additionally, $\mathcal{C}$ initializes an empty revocation table $\mathcal{R}$ to simulate the real system's attribute revocation state. This table will be used to answer the adversary's revocation queries consistently.

For a private key query on attribute set S. The challenge $\mathcal{C}$ randomly selects $r \in Z_q^*$, it finds a vector $\overrightarrow{\boldsymbol{\gamma}} = (\gamma_1, \cdots, \gamma_{n^*})$, such that $\gamma_1 = -1$ and $(\boldsymbol{M}_i^*)^\top \cdot \boldsymbol{\gamma} = 0$ (i is the row index corresponding to the attribute in attribute set S in matrix $\boldsymbol{M}^*$), If the set S of attributes does not match the access matrix $\boldsymbol{M}^*$, then such the vector $\overrightarrow{\boldsymbol{\gamma}}$ must exist, as detailed in Section 2 of [30]. $\mathcal{C}$ calculates

$$K' = g^t = g^{r + \sum_{i=1}^{n^*} \gamma_i \alpha^{q+1-i}} = g^r \prod_{i=1}^{n^*} \left(g^{\alpha^{q+1-i}} \right)^{\gamma_i}$$

$\mathcal{C}$ implicitly set

$$t = r + \gamma_1 \alpha^q + \gamma_2 \alpha^{q-1} + \cdots + \gamma_{n^*} \alpha^{q-n^*+1}$$

$\mathcal{C}$ calculates

$$K = g^s g^{\alpha t} = g^{\alpha r} \prod_{i=1}^{n^*} \left(g^{\alpha^{q+1-i}} \right)^{\gamma_i} \cdot g^{\alpha^{q+1}} \cdot g^{s'}$$

Because $\gamma_1 = -1$, so K can be reduced to

$$K = g^{s'} g^{\alpha r} \prod_{i=2}^{n^*} \left(g^{\alpha^{q+2-i}} \right)^{\gamma_i}$$

For a revocation query on tuple (c, UIM_d). $\mathcal{C}$ updates $\mathcal{R}[UIM_d][c] \leftarrow \texttt{revoked}$. Since $\mathcal{C}$ fully controls the simulation of the hash function H_3 and the mapping $\rho(i)$ between attributes and matrix rows, it can perfectly track which rows of any (simulated or challenge) ciphertext correspond to the attribute c. Therefore, $\mathcal{C}$ can consistently simulate the effect of the Attribute-Revoke algorithm by marking the corresponding ciphertext components (rows of M^* and pairs (C_i, C_i')) as invalid for the user UIM_d in its internal simulation state. This does not require $\mathcal{C}$ to perform any new cryptographic operation or alter the underlying reduction to the p-PBDHE assumption.

The challenger $\mathcal{C}$ must construct the ciphertext component K_c for each row of the access matrix $\boldsymbol{M}^*$. Let the attribute associated with row i be $\rho^*(i)$. We define $\eta_x = H_2(x)$ for an attribute x. There are two distinct cases:

Case 1: $\rho^*(i) \notin S$, where S is the attribute set submitted by the adversary in Phase 1. In this case, the attribute is not in the adversary's key, and $\mathcal{C}$ can simulate K_{c_i} directly using the random oracle H_1 and the known value $t = r + \sum_{j=1}^{n^*} \gamma_j \alpha^{q+1-j}$. Since $\mathcal{C}$ knows g^{α^j} for $j \in \{1, \ldots, q\}$ from the p-BDHE instance, it computes:

$$K_{c_i} = H_1(\rho^*(i))^t = (g^{\eta_{\rho^*(i)}})^t = g^{\eta_{\rho^*(i)} \cdot t}.$$

This value is correctly distributed as t is implicitly defined and $\eta_{\rho^*(i)}$ is a known random exponent chosen by $\mathcal{C}$ for the random oracle.

Case 2: $\rho^*(i) \in S$. This is the challenging case because the adversary holds a secret key for this attribute, and $\mathcal{C}$ must embed the hard problem without knowing the master secret α. Recall that the secret sharing vector is $\mathbf{v} = (s, v_2, \ldots, v_{n^*})^T = (s, \gamma_1 \alpha^q + \gamma_2 \alpha^{q-1} + \cdots + \gamma_{n^*} \alpha^{q-n^*+1})^T$, where $s = \gamma_1 \alpha^{q+1} + \gamma_2 \alpha^q + \cdots + \gamma_{n^*} \alpha^{q-n^*+2}$ is unknown to $\mathcal{C}$. For a row i with attribute $\rho^*(i) \in S$, the secret share is $\lambda_i = \boldsymbol{M}_i^* \cdot \mathbf{v}$. The corresponding K_{c_i} should be computed as $K_{c_i} = H_1(\rho^*(i))^{\lambda_i}$.

To simulate this, $\mathcal{C}$ leverages the condition that for any unauthorized set of rows I, there exists a vector $\mathbf{w}$ such that $\mathbf{w} \cdot \mathbf{M}_I^* = (1, 0, \ldots, 0)$ and $w_i = 0$ for $i \notin I$. In the simulation, $\mathcal{C}$ programs the random oracle H_1 for attribute $\rho^*(i)$ as $H_1(\rho^*(i)) = g^{b_i} \cdot g^{\eta_{\rho^*(i)}}$, where b_i is a known random exponent. Then, the computation unfolds as follows:

$$\begin{aligned} K_{c_i} &= H_1(\rho^*(i))^{\lambda_i} = \left(g^{b_i} \cdot g^{\eta_{\rho^*(i)}} \right)^{\lambda_i} \\ &= g^{b_i \lambda_i} \cdot g^{\eta_{\rho^*(i)} \lambda_i}. \end{aligned}$$

The term $g^{b_i \lambda_i}$ contains the unknown α^{q+1} because λ_i includes the component $\gamma_1 \alpha^{q+1}$ from s. However, by carefully choosing the vector $\gamma = (\gamma_1, \ldots, \gamma_{n^*})$ such that for all rows i with $\rho^*(i) \in S$, we have $\boldsymbol{M}_i^* \cdot \gamma = 0$, the terms involving α^{q+1} cancel out. Specifically, expanding $\lambda_i = \boldsymbol{M}_i^* \cdot \mathbf{v}$ and using the condition $\boldsymbol{M}_i^* \cdot \gamma = 0$, $\mathcal{C}$ can express $g^{b_i \lambda_i}$ using only the known terms from the p-BDHE instance (i.e., g^{α^j} for $j = 1, \ldots, 2q, j \neq q+1$). The detailed algebraic manipulation yields:

$$K_{c_i} = (g^{\eta_{\rho^*(i)}})^{\lambda_i} \cdot \prod_{j=1}^{n^*} \left(g^{b_i \gamma_j \alpha^{q+1-j}} \right)^{M^*_{i,j}}.$$

Since $\mathcal{C}$ knows $g^{\alpha^{q+1-j}}$ for all j (as $q+1-j \neq q+1$), it can compute the above product. Finally, K_c for the entire ciphertext is the set $\{K_{c_i}\}$ for all rows i. This simulation is perfect, and the adversary cannot distinguish it from a real ciphertext unless it can solve the p-BDHE problem.

Challenge. Finally, $\mathcal{A}$ sends two same length messages M_0, M_1 to $\mathcal{C}$, $\mathcal{C}$ selects $b \in \{0, 1\}$ to generate a challenge ciphertext for M_b. $\mathcal{C}$ selects a vector $\overrightarrow{\boldsymbol{v}} = (\zeta, y_2, y_3, \cdots, y_{n^*})$,then $\mathcal{C}$ creates $\widetilde{C} = m \cdot T \cdot e(g^\zeta, g^{s'})$, $C = g^\zeta$, where T is the challenge tuple generated during the initialization phase of the $\mathcal{C}$. But the simulation for C_i is difficult because we embed the hard problem in the attribute key, in the ciphertext to form a correspondence with the simulated key, and in the presence of the part of C_i that we want to eliminate, so the challenger $\mathcal{C}$ uses secret partitioning to simulate C_i. To eliminate some of the values, $\mathcal{C}$ changes the vector $\overrightarrow{\boldsymbol{v}}$ to $\overrightarrow{\boldsymbol{v}} = (\zeta, \zeta\alpha + y_2, \zeta\alpha^2 + y_3, \cdots, \zeta\alpha^{n^*-1} + y_{n^*})$,

$\mathcal{C}$ calculates $\delta_i = \boldsymbol{M}^* \cdot \overrightarrow{\boldsymbol{v}}$, where $i = 1, \cdots, n$.

First, $\mathcal{C}$ randomly selects $(\widetilde{r_1}, \widetilde{r_2}, \cdots, \widetilde{r_{n^*}})$ to generate $C_i^{'} = g^{-\widetilde{r_i}} g^{-\zeta b_i}$. $\mathcal{C}$ implicitly sets $r_i = -\widetilde{r_i} + \zeta b_i$. $\mathcal{C}$ calculates

$$C_i = g^{\alpha \delta_i} \cdot H_1(\rho^*(i))^{-r_i} (i = 1, 2, \cdots, n^*)$$

The challenger $\mathcal{C}$ define U_i to be a set of all other row indices with the same attributes as row i and brings in

$$H_1(x) = g^{\eta_x} \prod_{i \in B} g^{\alpha M_{i,1}^*/\beta_i} \cdot g^{\alpha^2 M_{i,2}^*/\beta_i} \cdots g^{\alpha^{l^*} M_{i,l^*}^*/\beta_i}$$

to simplify to get $C_{1,i} = H_1(\rho^*(i))^{\widetilde{r_i}} \left(\prod_{j=2}^{l^*} g^{a M_{i,j}^* y_i} \right)$ and $C_{2,i} = g^{-b_i \zeta \eta_{\rho_i^*}} \left(\prod_{k \in U_i} \prod_{j=1}^{l^*} \left(g^{a^i \zeta (b_i/b_k)} \right)^{M_{k,j}^*} \right)$. $\mathcal{C}$ calculates $C_i = C_{1,i} \cdot C_{2,i} (i = 1, 2, \cdots, n^*)$, and sends challenge ciphertext $CT^* = \left\{ (C_i, C_i^{'})_{i=1,2,\cdots,n^*}, C, \widetilde{C} \right\}$ to $\mathcal{A}$.

Query Phase 2. Same as ***Query Phase 1***.

Guess. The adversary $\mathcal{A}$ outputs a guess b', if $b' = b$, $\mathcal{C}$ outputs 0 to guess that $T = e(g, g)^{a^{q+1} s}$; Otherwise, $\mathcal{C}$ outputs 1 to guess that T is randomly element R.

When T is a tuple the simulator $\mathcal{C}$ gives a perfect simulation so we have that

$$\Pr \left[\mathcal{C} \left(\mathbb{D}, T = e(g, g)^{\alpha^{q+1} \zeta} \right) = 0 \right] = \tfrac{1}{2} + \mathsf{Adv}_{\mathcal{A}}(\kappa).$$

If T is a random tuple, then the message M_b is completely hidden from the adversary $\mathcal{A}$, and we have $\Pr \left[\mathcal{C}(\mathbb{D}, T \in G_T) = 0 \right] = \frac{1}{2}$. Therefore, if the advantage $\mathsf{Adv}_{\mathcal{A}}(\kappa)$ of $\mathcal{A}$ in the above game is not negligible, then $\mathcal{A}$ breaks the selective CPA security of our scheme, that is, $\mathcal{C}$ has non-negligible advantage in solving p-PBDH assumption. □

C. Security Analysis

Next, we will analyze the relevant security properties of our CP-ABE scheme. Furthermore, we provide a critical discussion on the practical implications of the core cryptographic assumptions underlying our design within the VANET context.

(1) ***Anonymity***. We propose an attribute-based solution for VANET's data engineering that utilizes only user attributes as private information. In our scheme, a cryptographic hash function H_3 conceals user attributes during decryption, enabling decryptors to privately verify attribute compliance with the access policy without revealing actual attributes to the server, thereby preserving decryptor anonymity. The scheme's anonymity is fundamentally guaranteed by the one-way property of the hash function. During encryption, vehicles employ access policies for data protection, while the authority center obscures policy information through the hash function H_2 and an optimized hash chain table, preventing policy structure reconstruction by adversaries. This approach simultaneously ensures the encryptor's anonymity and security, both of which rely on the irreversible nature of hash functions.

(2) ***Collusion Resistant***. Collusion resistance refers to a system's capability to withstand coordinated attempts by multiple entities to collectively exploit their resources or privileges to subvert its normal operation, fairness, or integrity. In cryptographic protocols, collusion resistance ensures that even when multiple adversaries collaborate and pool their resources, they remain unable to breach the security or privacy guarantees of the scheme. In our system's user registration phase, the TA generates a unique random value $t \in Z_q^*$ for each user to derive their attribute-specific keys. Since each user's t is distinct, malicious users cannot combine their keys to launch collusion attacks, thereby preserving the system's security.

(3) ***Revocability***. Attribute revocation refers to the capability of a system to dynamically withdraw specific attributes or permissions from a user when required. In our scheme, this mechanism ensures that revoked attributes become invalid for ciphertext decryption. Specifically, our revocation algorithm mandates that when an attribute is revoked, the TA removes the corresponding ciphertext component associated with that attribute. As a result, the user's decryption key no longer matches the modified ciphertext, rendering successful decryption impossible.

(4) ***Forward and Backward Security***. Forward security ensures that past cryptographic keys remain secure even if an adversary compromises the current key. In our scheme, users must register and obtain their decryption keys from the TA, which generates attribute-specific keys using the master secrets s and α. Since α remains confidential to the TA and its derivation from the public key g^α is computationally infeasible under the computational Diffie-Hellman assumption, users cannot recover the master key. Furthermore, the TA incorporates a unique random value t in each key generation, ensuring distinct attribute keys across different stages. Although our scheme does not explicitly incorporate temporal factors, the revocation mechanism guarantees security upon attribute revocation: the TA removes the corresponding ciphertext component, invalidating the user's key for that attribute and preventing successful decryption. Backward security similarly ensures the protection of future keys even if the current key is exposed. The same principles outlined above apply to backward security in our scheme.

(5) ***Practical Implications of Security Assumptions***. The formal security of our scheme rests on the decisional p-PBDHE assumption within the Random Oracle Model (ROM), under a selective security game. It is crucial to examine the practical impact of these choices for VANET deployments. The selective security model, where the adversary must declare the challenge access structure upfront, is a standard and accepted model for proving the security of expressive ABE systems. In VANETs, many access policies (e.g., "police vehicle in zone 5") are defined by operational roles or event types, which are often determined independently of the specific cryptographic system parameters, making selective security a pragmatically viable model. Nevertheless, achieving full adaptive security, where the challenge policy can be chosen adaptively, remains a valuable theoretical goal for future work. The reliance on the ROM implies that our security proofs hold assuming the employed hash functions (H_1, H_2, H_3) behave as ideal random functions. In practice, they are instantiated with standardized, collision-resistant hash functions (e.g., from the SHA-3 family). While no concrete function is a perfect random oracle, the ROM is a well-vetted heuristic underlying the security of numerous deployed protocols, allowing us to achieve the efficiency required for resource-constrained vehic-

TABLE VI
FUNCTIONAL FEATURE COMPARISON OF CP-ABE SCHEMES

Scheme	Policy Hidden	Large Universe	Attribute Revoke	Outsourcing Decryption
Feng *et al.* [13]	×	×	×	√
Li *et al.* [23]	×	×	×	√
Li *et al.* [25]	Partially	×	×	×
Xiong *et al.* [33]	Partially	√	√	×
Zhang *et al.* [40]	Partially	√	√	×
Yang *et al.* [36]	Partially	×	√	√
Ruan *et al.* [28]	Partially	√	×	√
Our Scheme	**Fully**	√	√	√

ular networks. Finally, the p-PBDHE assumption necessitates the use of a secure, pairing-friendly elliptic curve group (e.g., BN or BLS12 curves) whose parameters are chosen to provide sufficient security strength against known attacks, aligning with standard practices for pairing-based cryptography.

D. Performance Comparison

A comparative analysis of our construction with existing ABE schemes is conducted from two perspectives: functional completeness and performance efficiency. First, a high-level feature comparison is presented in Table VI, evaluating key attributes including policy hiding, large universe support, attribute revocation, and outsourced decryption. The literature review reveals that no single existing scheme integrates all four features. For instance, while [33] and [40] support policy hiding, revocation, and large universes, they lack outsourced decryption. Conversely, schemes like [23] and [13] implement outsourced decryption but omit policy hiding and revocation. The schemes by [36] and [28] represent advanced designs that combine outsourcing with either revocation or large universe support, yet still achieve only partial policy hiding. This analysis underscores a clear gap: our proposed scheme is the first to simultaneously offer *full* policy hiding, large universe, attribute revocation, and secure outsourced decryption within a unified framework.

For the performance evaluation in the following sections, we select a representative set of schemes that provide meaningful benchmarks across different performance dimensions. We include Xiong *et al.* [33] and Zhang *et al.* [40] as the primary functional competitors, as they share the closest feature set (partial hiding + revocation). To benchmark computational and communication efficiency in specific areas, we also compare against: Yang *et al.* [36], notable for its integration of outsourcing with revocation; and Ruan *et al.* [28], recognized for its compact ciphertext structure and efficient outsourcing. This selection allows us to demonstrate that our scheme not only integrates more features but also remains highly competitive or superior in terms of computational speed, storage, and communication overhead when compared to state-of-the-art schemes optimized for specific performance goals.

VII. EFFICIENCY ANALYSIS OF OUR CONSTRUCTION

To validate the practical applicability and computational efficiency of our proposed scheme, we conduct comprehensive performance simulations and comparative analyses with existing approaches. In accordance with the comparative framework established in Section I-A, we benchmark our scheme against four state-of-the-art and representative CP-ABE constructions: Xiong *et al.*'s scheme [33] and Zhang *et al.*'s scheme [40], as the most direct functional competitors supporting policy hiding and revocation; Yang *et al.*'s scheme [36], noted for its integration of outsourcing with revocation; and Ruan *et al.*'s scheme [28], recognized for its compact ciphertext structure. This selection allows a rigorous, multi-dimensional evaluation of whether our scheme, which integrates a broader set of features (full hiding, revocation, outsourcing), can also achieve competitive or superior performance in terms of computational overhead, storage footprint, and communication cost.

We employ an elliptic curve defined over a higher-order cyclic group, specified by the equation $E : y^2 = x^3 + 3x \mod q$, where q is a 512-bit prime. The group G constitutes a cyclic subgroup of E with prime order q (160 bits). Our construction utilizes a bilinear pairing $e : G \times G \rightarrow G_T$, where $|G| = |G_T| = 512$ bits (128 bytes) and $|Z_q^*| = 160$ bits (20 bytes). The computational operations referenced in our performance analysis are formally defined in Table VII.

TABLE VII
DEFINITION OF SYMBOLS IN PERFORMANCE ANALYSIS

Symbol	Definition
$\|G\|, \|G_T\|, \|Z_q^*\|$	The size of an element in G, G_T, Z_q^*
$\|U\|, \|C\|$	The number of users/attributes in the system
n	The number of rows of the access matrix $\boldsymbol{M_{n\times l}}$
S	An attribute set that satisfies the access policy
n^*	The size of the minimal satisfying attribute set S

Our experimental evaluation employs the MIRACL cryptographic library [1] to benchmark algorithm performance across heterogeneous hardware platforms. The testbed configuration comprises: (1) *Embedded Platform*. Raspberry Pi with 32-bit dual-core ARM Cortex-A7 and Cortex-M4 processors (FPU/MPU enabled); 1GB RAM running Linux OS. (2) *Development Environment*. MacBook Air (M1 processor, 8GB RAM) running macOS 13.0; Windows 11 emulation via Parallels Desktop; SSH connectivity to Raspberry Pi for remote execution. The computational benchmarks for core cryptographic operations, presented in Table VIII, were obtained through

[1] https://github.com/miracl/MIRACL

TABLE IX
COMPUTATIONAL OVERHEAD

Scheme	Xiong *et al.* [33]	Yang *et al.* [36]	Zhang *et al.* [40]	Xiong *et al.* [28]	Our Scheme
Attribute-Key Generation	$(8\|C\|+5)T_G^e$	$(\|C\|+4)T_G^e$	$(2+\|C\|)T_G^e$	$(4+2\|C\|)T_G^e$	$(2+\|C\|)T_G^e$
Encryption	$(4+8n)T_G^e+T_p+T_{G_T}^e$	$(5n+3)T_G^e+T_{G_T}^e$	$(6n+2)T_G^e+2T_{G_T}^e$	$(3n+1)T_G^e+T_{G_T}^e$	$(3n+1)T_G^e+T_{G_T}^e$
Decryption	$(4+6n^*)T_p$	$(3n^*+3)T_p+(n^*+2)T_G^e$	$(2+2n^*)T_p$	$(1+2n^*)T_p+2T_G^e$	T_p

this experimental setup. For reproducibility and verification purposes, we provide a supplementary video demonstration [2] detailing the complete simulation process.

TABLE VIII
THE COMPUTATION COST OF THE UNDERLYING OPERATIONS

Symbol	Algorithm	Computation Cost (ms)
T_p	Bilinear Pairing operation	15.9218
T_G^e	Exponentiation in group G	6.7006
$T_{G_T}^e$	Exponentiation in group G_T	1.7253

To rigorously evaluate our scheme's advantage, we conduct a comprehensive comparative analysis against the selected benchmark schemes [28], [33], [36], [40]. Using execution time metrics obtained from our Raspberry Pi testbed, we systematically compare: (1) *Computational Efficiency*. Encryption/decryption latency and key generation overhead across varying parameters. (2) *Storage Requirements*. Public parameter, key, and ciphertext sizes. (3) *Communication Overhead*. Bandwidth consumption for ciphertext upload. The experimental results demonstrate that our construction not only integrates more security features but also maintains highly competitive, and often superior, performance across all these metrics compared to existing schemes that excel in only one or two dimensions.

A. Computational Overhead Comparison

For systematic performance evaluation, we categorize the computational overhead into three distinct components: Attribute Key Generation, Encryption Process, and Decryption Process. The quantitative analysis of these computational costs, as measured through our Raspberry Pi testbed, is presented in Table IX. Furthermore, we provide visual comparisons of the respective timing metrics in Fig. 4 (key generation), Fig. 5 (encryption), and Fig. 6 (decryption), enabling clear performance benchmarking across different operational phases.

By integrating the results from Fig. 4 and Table IX with a fixed attribute set size ($|C| = 5$), our analysis reveals several key findings regarding computational efficiency. In key generation, our scheme achieves efficiency comparable to the most efficient schemes, Zhang *et al.*'s and Ruan *et al.*'s, both having a complexity of $(2+|C|)T_G^e$. Yang *et al.*'s scheme is also efficient with $(|C|+4)T_G^e$, while Xiong *et al.*'s scheme exhibits a significantly higher overhead of $(8|C|+5)T_G^e$.

For encryption efficiency with a fixed policy size ($n = 5$), Fig. 5 and Table IX demonstrate a clear hierarchy. Xiong *et al.*'s scheme incurs the highest overhead due to its $(4+8n)$

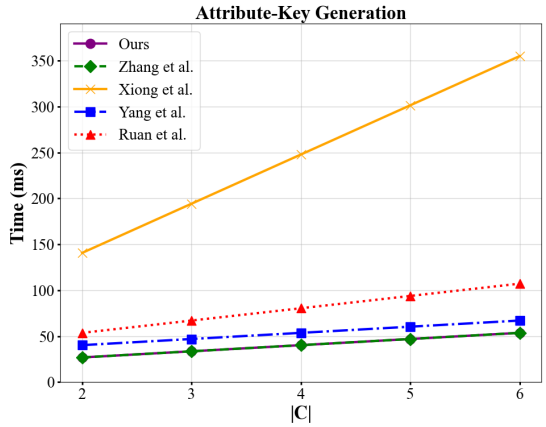

Fig. 4. Computational overheads in key generation

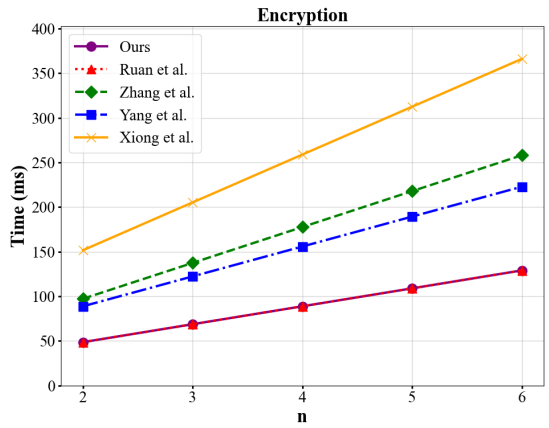

Fig. 5. Computational overheads in Encrypt

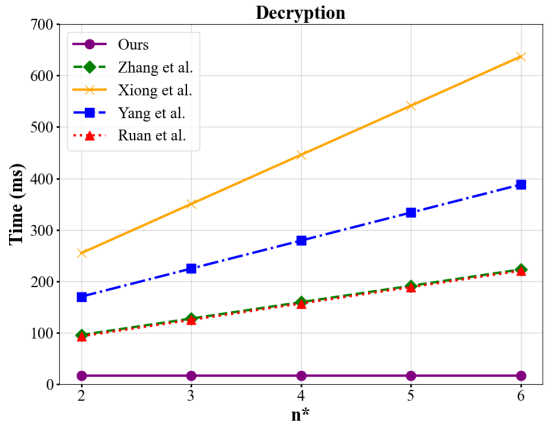

Fig. 6. Computational overheads in Decrypt

[2]https://github.com/mkllsr/Revocation-algorithm-time-and-Video-of-Raspberry-Pi-in-action

factor. Zhang *et al.*'s scheme shows moderate efficiency. Crucially, our scheme achieves the same high efficiency as Ruan *et al.*'s scheme, both with a low $(3n+1)$ multiplicative factor, outperforming Yang *et al.*'s scheme which has a $(5n+3)$ factor. This advantage stems from our optimized cryptographic operations during the encryption phase.

The decryption comparison $(n^* = 2)$ in Fig. 6 highlights the most significant performance differentiator. Schemes without full outsourcing, namely Xiong and Zhang, require multiple pairing operations, leading to substantial and policy-dependent latency. Yang *et al.*'s scheme, while supporting outsourcing, still requires the user to perform operations scaling with n^*. Ruan *et al.*'s scheme offers more efficient outsourcing but retains a linear component. In stark contrast, our scheme achieves optimal, constant-time performance on the user side (T_p) through the complete offloading of all pairing operations to the cloud. This strategic outsourcing, combined with server-assisted mechanisms, minimizes the local computational burden to a single, lightweight operation, providing a decisive advantage for resource-constrained devices.

To further evaluate the practical applicability in large-scale VANETs where the number of attributes $(|C|)$ and policy complexity (n) can be substantial, we analyze the theoretical end-to-end latency, comprising key generation, encryption, and decryption, as the system scales. The analysis assumes a typical scenario where the access policy size n grows proportionally to the attribute set size $|C|$, and a user requires half of the attributes $(n^* = |C|/2)$ for decryption. The theoretical latency for each scheme is derived based on the complexity formulas in Table IX and the operation costs in Table VIII.

Fig. 7 illustrates the theoretical end-to-end latency across a wide range of attribute counts ($|C|$ from 10 to 200). A critical observation is that all schemes exhibit linear growth $O(|C|)$ in latency, as expected from their algebraic formulations. However, the slopes differ significantly. Our scheme, along with Ruan *et al.*'s, demonstrates the flattest growth curve, confirming the efficiency of their encryption design. More importantly, the gap between our scheme and others widens as $|C|$ increases, primarily due to our offloading of all pairing operations during decryption. While other schemes, including the efficient Ruan *et al.*, incur linearly increasing costs from local pairing operations, our scheme confines the user's local decryption to a single, constant-time exponentiation. This result robustly demonstrates that our integrated approach not only enhances security with full policy hiding and revocation but also delivers superior and predictable performance essential for dynamic, large-scale VANET deployments.

B. Storage Overhead Comparison

Our storage overhead analysis systematically evaluates three critical components: Public Parameters, Attribute Keys, and Ciphertexts. The quantitative storage requirements, measured against pre-defined standard group sizes, are presented in Table XI. For visual comparison, we provide Fig. 8to show public parameter storage overhead, Fig. 9 to show attribute key storage requirements, and Fig. 10 to show Ciphertext storage footprint.

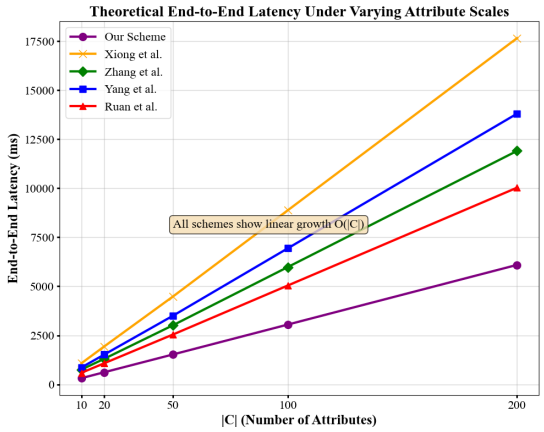

Fig. 7. Theoretical end-to-end latency (key generation, encryption, and decryption) versus the number of attributes $|C|$. Our scheme demonstrates the most scalable performance, maintaining the lowest latency due to its efficient encryption and outsourced decryption.

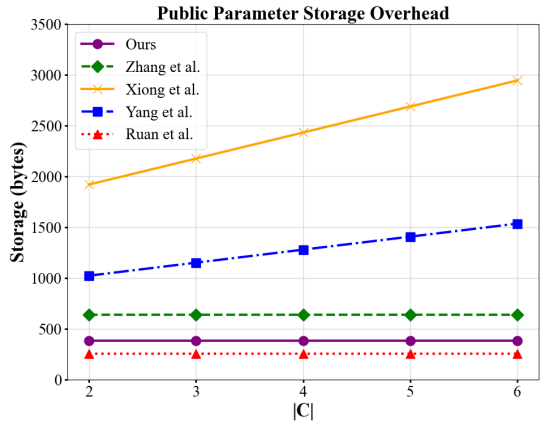

Fig. 8. Storage overheads of public parameter

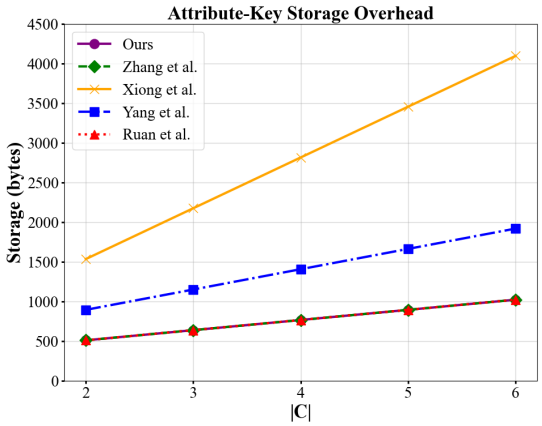

Fig. 9. Storage overheads of private key

Since the public parameter component in [33] incorporates $|U|$, whereas the other comparative studies do not, we standardized $|U| = 2$ for a fair benchmarking. As illustrated in Fig. 8 and Table XI, the storage overhead for public parameters varies significantly. Xiong *et al.*'s scheme has the largest size, which grows with the number of system users $|U|$. Zhang

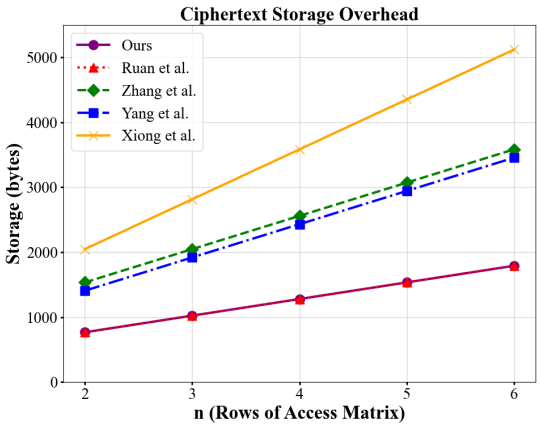

Fig. 10. Storage overheads of Ciphertext

et al.'s, Yang *et al.*'s, and our scheme have constant-sized or user-independent parameters, with our scheme achieving the most compact size of $2|G| + |G_T|$. Notably, Ruan *et al.*'s scheme also boasts a very small, constant-sized public parameter, denoted as $O(1)|G| + |G_T|$.

Regarding attribute key storage requirements, Table XI and Fig. 9 reveal a linear relationship between key size and the number of user attributes $|C|$. Under $|C| = 5$, our scheme exhibits the same efficient growth rate $(2 + |C|)|G|$ as Zhang *et al.*'s and Ruan *et al.*'s schemes. Xiong *et al.*'s scheme shows a much steeper slope $(2+5|C|)|G|$, while Yang *et al.*'s scheme has a moderate overhead of $(3 + 2|C|)|G|$. This confirms the efficiency of our key structure.

For ciphertext dimensions, Table XI and Fig. 10 demonstrate a linear correlation with the access matrix row count n. With $n = 5$ for standardized comparison, our solution and Ruan *et al.*'s scheme achieve the minimal and identical expansion rate of $(2n + 1)|G| + |G_T|$, establishing a clear advantage in ciphertext space efficiency. Xiong *et al.*'s scheme produces the largest ciphertexts, while Zhang *et al.*'s and Yang *et al.*'s schemes lie in between, with Zhang's being slightly larger due to an extra $|G_T|$ element.

Beyond computational and storage efficiency, communication overhead is a critical performance metric for VANETs, as it directly impacts bandwidth consumption during data dissemination (e.g., from vehicles to the cloud). This section provides a theoretical and empirical analysis of the communication cost for ciphertext upload, which is typically the most voluminous and frequent data transfer in CP-ABE systems. We compare our scheme with [28], [33], [36], [40]. The analysis is based on the cryptographic parameters defined in Table VII: $|G| = 128$ bytes and $|G_T| = 128$ bytes. The primary variable is n, the number of rows in the LSSS access matrix, representing the policy complexity.

The communication cost for uploading a ciphertext is dominated by the number of group elements (G and G_T) it contains. The theoretical formulas for the compared schemes are summarized in Table X.

As shown in Table X, the ciphertext size in all schemes grows linearly with the policy complexity n. The key difference lies in the slope. Xiong *et al.*'s scheme has the steepest slope (coefficient 6 for n), indicating the highest bandwidth consumption. Zhang *et al.*'s and Yang *et al.*'s schemes have identical slopes (coefficient 4) for the $|G|$ components, but Zhang's scheme includes an additional $|G_T|$ element. Our scheme and Ruan *et al.*'s scheme achieve the most compact representation, sharing the smallest slope (coefficient 2). This design is crucial for bandwidth-constrained environments.

TABLE X
THEORETICAL CIPHERTEXT UPLOAD OVERHEAD

Scheme	Ciphertext Size (Bytes)
Xiong *et al.* [33]	$(6n + 3)\|G\| + \|G_T\|$
Zhang *et al.* [40]	$(4n + 2)\|G\| + 2\|G_T\|$
Yang *et al.* [36]	$(4n + 2)\|G\| + \|G_T\|$
Ruan *et al.* [28]	$(2n + 1)\|G\| + \|G_T\|$
Our Scheme	$(2n + 1)\|G\| + \|G_T\|$

Fig. 11 clearly illustrates these theoretical differences. For any given n, the ciphertext size of our scheme and Ruan *et al.*'s is the smallest and identical. The performance gap between this most efficient group and the others widens significantly as the policy becomes more complex. For instance, with $n = 20$, our scheme produces a ciphertext of approximately 5,000 bytes, while Xiong *et al.*'s scheme results in nearly 16,000 bytes—over three times larger. Yang's and Zhang's schemes produce intermediate-sized ciphertexts. This trend confirms that the advantage of compact ciphertext structures like ours and Ruan's becomes more pronounced in scenarios requiring expressive access policies, which are common in VANETs.

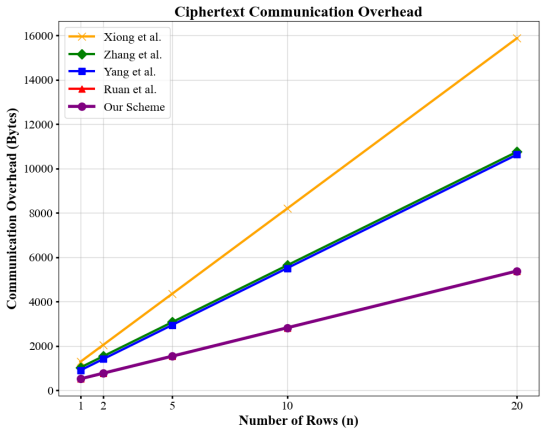

Fig. 11. Ciphertext Communication Overhead for different schemes as a function of the access policy rows (n). Our scheme demonstrates the most efficient and scalable communication cost.

Fig. 11 clearly illustrates the superior efficiency of our scheme. For any given n, the ciphertext size of our scheme (and Ruan *et al.*'s) is the smallest. Crucially, as the policy becomes more complex (i.e., n increases), the performance gap widens significantly. For instance, with $n = 20$, our scheme produces a ciphertext of approximately 5,000 bytes, while Xiong *et al.*'s scheme results in nearly 16,000 bytes—over three times larger. This trend confirms that the advantage of our compact ciphertext structure becomes more pronounced in scenarios requiring expressive access policies, which are common in VANETs.

TABLE XI
STORAGE OVERHEADS

Scheme	Xiong *et al.* [33]	Zhang *et al.* [40]	Yang *et al.* [36]	Ruan *et al.* [28]	Our Scheme
Public Parameter	$(11+2\|U\|)\|G\|$	$4\|G\|+\|G_T\|$	$(5+\|C\|)\|G\|+\|G_T\|$	$O(1)\|G\|+\|G_T\|$	$2\|G\|+\|G_T\|$
Attribute-Key	$(2+5\|C\|)\|G\|$	$(2+\|C\|)\|G\|$	$(3+2\|C\|)\|G\|$	$(2+\|S\|)\|G\|$	$(2+\|C\|)\|G\|$
Ciphertext	$(6n+3)\|G\|+\|G_T\|$	$(4n+2)\|G\|+2\|G_T\|$	$(4n+2)\|G\|+\|G_T\|$	$(2n+1)\|G\|+\|G_T\|$	$(2n+1)\|G\|+\|G_T\|$

The analysis confirms that our scheme achieves the most efficient ciphertext upload communication overhead among the compared schemes. This low and linearly growing cost is crucial for VANETs, where bandwidth is a constrained resource and vehicles frequently broadcast or upload encrypted data (e.g., safety messages, sensor data). The minimized ciphertext size reduces channel congestion, decreases transmission latency, and improves the overall scalability of the secure communication system.

VIII. SCALABILITY ANALYSIS OF THE REVOCATION ALGORITHM

This section presents a thorough scalability analysis of our revocation algorithm, directly addressing the performance demands of dynamic VANET environments where access policies can be complex and revocation events may occur frequently. While our revocation mechanism maintains the core efficiency of the established row-removal approach [40], its operation on a *fully hidden* policy necessitates a dedicated evaluation of its overhead as system parameters scale. We provide both theoretical complexity analysis and empirical measurements to demonstrate that our algorithm remains lightweight and practical under the stressful conditions typical of large-scale, evolving VANETs.

The proposed revocation algorithm (Algorithm 1) performs a linear scan through the n rows of the access matrix $M_{n\times l}$. For each row, it checks a revocation list (a constant-time operation) and conditionally removes a fixed-size ciphertext component pair $\{C_i, C_i'\}$. Therefore, its time complexity is $O(n)$, or more precisely $O(r)$ where $r \leq n$ is the number of rows associated with the revoked attributes. This efficiency stems from avoiding cryptographic re-computations. The communication overhead for an update is the size of the updated ciphertext CT_f sent from the TA to the cloud: $|CT_f| = (2n_f+1)|G|+|G_T|$, where $n_f = n-r$. Notably, this overhead decreases linearly with the number of revoked rows r.

To validate the theoretical analysis, we conducted experiments to measure the algorithm's performance. The primary factor affecting revocation time is the policy size n. We measured the execution time of a single revocation operation for matrices of varying sizes. The results, plotted in Fig. 12, confirm a clear linear relationship between revocation time and n, validating the $O(n)$ complexity. For example, with $n = 100$, the revocation completes within 4 ms, ensuring rapid response even with complex policies.

Additionally, we evaluated the communication overhead of the updated ciphertext CT_f. As shown in Fig. 13, the size of CT_f decreases linearly as the number of revoked rows r increases, which is consistent with the theoretical formula

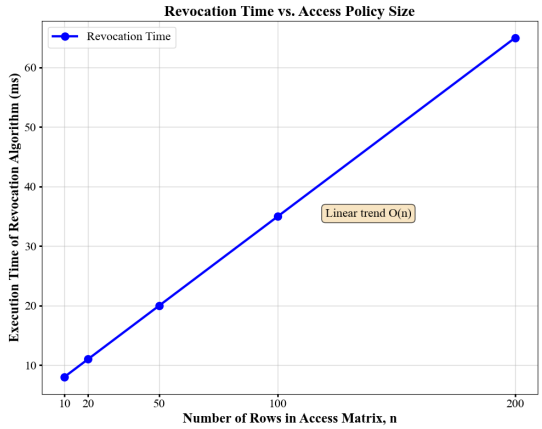

Fig. 12. Revocation time vs. policy size (n). The linear trend validates the $O(n)$ time complexity.

$|CT_f| = (2(n-r)+1)|G|+|G_T|$. This demonstrates that our revocation mechanism is bandwidth-efficient, especially when multiple attributes are revoked.

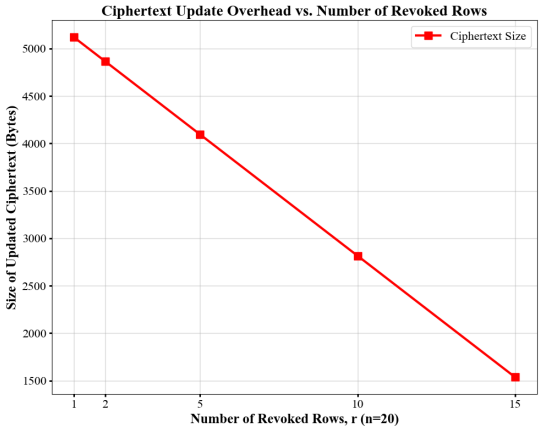

Fig. 13. Updated ciphertext size $|CT_f|$ vs. number of revoked rows (r), with initial $n = 20$. The overhead decreases linearly as more rows are removed.

For high-frequency revocation scenarios, the system handles each event as an independent transaction. The per-event cost is characterized by the time complexity shown in Fig. 12, and the primary workload falls on the well-provisioned TA. In scenarios with bursts of requests, the TA can employ batch processing to optimize throughput without altering the per-event computational complexity, ensuring system resilience.

The analysis confirms that our revocation mechanism is engineered for scalability. The linear-time complexity and decreasing communication overhead provide predictable performance as the system grows. This is crucial for VANETs, where policy complexity may increase and revocation frequency may

spike. The algorithm's efficiency ensures that security policy updates do not become a bottleneck, maintaining the system's agility and responsiveness, a fundamental requirement for securing dynamic vehicular networks.

IX. Conclusions

Our proposed system demonstrates the feasibility of employing attribute-based encryption (ABE) to strengthen the security and privacy of vehicular data in VANET environments. The solution delivers a scalable, efficient, and secure framework that enables flexible and fully hidden access control,dynamic revocation mechanisms, and secure outsourced decryption for VANET. We anticipate that this work will advance research in secure VANET and serve as a foundation for future investigations in this domain.

Several promising directions for further research : integration of fuzzy search over encrypted data, optimization of attribute-based access control for improved efficiency and accuracy, and scalability improvements to support larger vehicular. These extensions would further solidify the system's robustness and broaden its applicability in real-world deployments.

References

[1] Alaa Awad Abdellatif, Naram Mhaisen, Zina Chkirbene, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Reinforcement learning for intelligent healthcare systems: A comprehensive survey. *arXiv preprint arXiv:2108.04087*, 2021.

[2] Ruqayah R Al-Dahhan, Qi Shi, Gyu Myoung Lee, and Kashif Kifayat. Survey on revocation in ciphertext-policy attribute-based encryption. *Sensors*, 19(7):1695, 2019.

[3] Mohammad Ali, Mohammad-Reza Sadeghi, and Ximeng Liu. Lightweight revocable hierarchical attribute-based encryption for internet of things. *IEEE Access*, 8:23951–23964, 2020.

[4] Yang Ba, Xuexian Hu, Yue Chen, Zenghang Hao, Xuewei Li, and Xincheng Yan. A blockchain-based cp-abe scheme with partially hidden access structures. *Security and Communication Networks*, 2021:1–16, 2021.

[5] John Bethencourt, Amit Sahai, and Brent Waters. Ciphertext-policy attribute-based encryption. In *2007 IEEE symposium on security and privacy (SP'07)*, pages 321–334. IEEE, 2007.

[6] Melissa Chase. Multi-authority attribute based encryption. In *Theory of Cryptography: 4th Theory of Cryptography Conference, TCC 2007, Amsterdam, The Netherlands, February 21-24, 2007. Proceedings 4*, pages 515–534. Springer, 2007.

[7] Liqing Chen, Yu Liu, Hao Zhang, and Jian Weng. Multi-authority attribute-based searchable encryption with attribute revocation for internet of vehicles. *IEEE Transactions on Vehicular Technology*, pages 1–12, 2025.

[8] Liqing Chen, Zhile Tai, Hao Zhang, Jian Weng, and Zhiquan Liu. Traceable and revocable multi-authority attribute-based encryption with cloud and fog computing feasible for iov. *IEEE Transactions on Dependable and Secure Computing*, pages 1–14, 2026.

[9] Liqing Chen, Shiguo Xu, Chunhua Jin, Hao Zhang, and Jian Weng. Partially hidden policy attribute-based multi-keyword searchable encryption with verification and revocation. *IEEE Transactions on Mobile Computing*, 24:9020–9035, 2025.

[10] P Chinnasamy, P Deepalakshmi, Ashit Kumar Dutta, Jinsang You, and Gyanendra Prasad Joshi. Ciphertext-policy attribute-based encryption for cloud storage: Toward data privacy and authentication in ai-enabled iot system. *Mathematics*, 10(1):68, 2021.

[11] Hui Cui, Zhiguo Wan, Xinlei Wei, Surya Nepal, and Xun Yi. Pay as you decrypt: Decryption outsourcing for functional encryption using blockchain. *IEEE Transactions on Information Forensics and Security*, 15:3227–3238, 2020.

[12] Sangjukta Das and Suyel Namasudra. Macpabe: Multi-authority-based cp-abe with efficient attribute revocation for iot-enabled healthcare infrastructure. *International journal of network management*, 33(3):e2200, 2023.

[13] Chaosheng Feng, Keping Yu, Moayad Aloqaily, Mamoun Alazab, Zhihan Lv, and Shahid Mumtaz. Attribute-based encryption with parallel outsourced decryption for edge intelligent iov. *IEEE Transactions on Vehicular Technology*, 69(11):13784–13795, 2020.

[14] Tingyun Gan, Yongjian Liao, Yikuan Liang, Zijun Zhou, and Ganglin Zhang. Partial policy hiding attribute-based encryption in vehicular fog computing. *Soft Computing*, 25:10543–10559, 2021.

[15] Sheng Gao, Guirong Piao, Jianming Zhu, Xindi Ma, and Jianfeng Ma. Trustaccess: A trustworthy secure ciphertext-policy and attribute hiding access control scheme based on blockchain. *IEEE Transactions on Vehicular Technology*, 69(6):5784–5798, 2020.

[16] Chunpeng Ge, Willy Susilo, Joonsang Baek, Zhe Liu, Jinyue Xia, and Liming Fang. Revocable attribute-based encryption with data integrity in clouds. *IEEE Transactions on Dependable and Secure Computing*, 19(5):2864–2872, 2021.

[17] Chenghao Gu, Jiguo Li, Yichen Zhang, Yang Lu, and Jian Shen. Eabe-pufph: Efficient attribute-based encryption with reliable policy updating under full policy hiding. *IEEE Transactions on Computers*, 74(11):3750–3762, 2025.

[18] Dezhi Han, Nannan Pan, and Kuan-Ching Li. A traceable and revocable ciphertext-policy attribute-based encryption scheme based on privacy protection. *IEEE Transactions on Dependable and Secure Computing*, 19(1):316–327, 2022.

[19] Jialu Hao, Cheng Huang, Jianbing Ni, Hong Rong, Ming Xian, and Xuemin Sherman Shen. Fine-grained data access control with attribute-hiding policy for cloud-based iot. *Computer Networks*, 153:1–10, 2019.

[20] Xin Huang, Hu Xiong, Jinhao Chen, and Minghao Yang. Efficient revocable storage attribute-based encryption with arithmetic span programs in cloud-assisted internet of things. *IEEE Transactions on Cloud Computing*, 2021.

[21] Fawad Khan, Hui Li, Liangxuan Zhang, and Jian Shen. An expressive hidden access policy cp-abe. In *2017 IEEE Second International Conference on Data Science in Cyberspace (DSC)*, pages 178–186. IEEE, 2017.

[22] Allison Lewko and Brent Waters. Decentralizing attribute-based encryption. In *Annual international conference on the theory and applications of cryptographic techniques*, pages 568–588. Springer, 2011.

[23] Hang Li, Keping Yu, Bin Liu, Chaosheng Feng, Zhiguang Qin, and Gautam Srivastava. An efficient ciphertext-policy weighted attribute-based encryption for the internet of health things. *IEEE Journal of Biomedical and Health Informatics*, 26(5):1949–1960, 2021.

[24] Jiguo Li, Shaobo Chen, Yang Lu, Jianting Ning, Jian Shen, and Yichen Zhang. Revocable registered attribute-based encryption with user deregistration. *IEEE Internet of Things Journal*, 12(15):31526–31535, 2025.

[25] Jiguo Li, Yichen Zhang, Jianting Ning, Xinyi Huang, Geong Sen Poh, and Debang Wang. Attribute based encryption with privacy protection and accountability for cloudiot. *IEEE Transactions on Cloud Computing*, 10(2):762–773, 2020.

[26] Yongjian Liao, Ganglin Zhang, and Hongjie Chen. Cost-efficient outsourced decryption of attribute-based encryption schemes for both users and cloud server in green cloud computing. *IEEE Access*, 8:20862–20869, 2020.

[27] Suhui Liu, Jiguo Yu, Chunqiang Hu, and Mengmeng Li. Traceable multiauthority attribute-based encryption with outsourced decryption and hidden policy for ciot. *Wireless Communications and Mobile Computing*, 2021:1–16, 2021.

[28] Conghao Ruan, Chunqiang Hu, Ruifeng Zhao, Zewei Liu, Hongyu Huang, and Jiguo Yu. A policy-hiding attribute-based access control scheme in decentralized trust management. *IEEE Internet Things J.*, 10(20):17656–17665, 2023.

[29] Kamalakanta Sethi, Ankit Pradhan, and Padmalochan Bera. Practical traceable multi-authority cp-abe with outsourcing decryption and access policy updation. *Journal of Information Security and applications*, 51:102435, 2020.

[30] Brent Waters. Ciphertext-policy attribute-based encryption: An expressive, efficient, and provably secure realization. In *Public Key Cryptography–PKC 2011: 14th International Conference on Practice and Theory in Public Key Cryptography, Taormina, Italy, March 6-9, 2011. Proceedings 14*, pages 53–70. Springer, 2011.

[31] Axin Wu, Yinghui Zhang, Xiaokun Zheng, Rui Guo, Qinglan Zhao, and Dong Zheng. Efficient and privacy-preserving traceable attribute-based encryption in blockchain. *Annals of Telecommunications*, 74:401–411, 2019.

[32] Meiyan Xiao, Hongbo Li, Qiong Huang, Shui Yu, and Willy Susilo. Attribute-based hierarchical access control with extendable policy. *IEEE*

Transactions on Information Forensics and Security, 17:1868–1883, 2022.

[33] Hu Xiong, Yanan Zhao, Li Peng, Hao Zhang, and Kuo-Hui Yeh. Partially policy-hidden attribute-based broadcast encryption with secure delegation in edge computing. *Future Generation Computer Systems*, 97:453–461, 2019.

[34] Shengmin Xu, Guomin Yang, and Yi Mu. Revocable attribute-based encryption with decryption key exposure resistance and ciphertext delegation. *Information Sciences*, 479:116–134, 2019.

[35] Xixi Yan, Guanghui He, Jinxia Yu, Yongli Tang, and Mingjie Zhao. Offline/online outsourced attribute-based encryption with partial policy hidden for the internet of things. *Journal of Sensors*, 2020:1–11, 2020.

[36] Xiaodong Yang, Chengeng Wang, Ruiting Li, and Chen Chen. A verifiable CP-ABE scheme with policy hiding and dynamic attribute revocation for secure medical iot system. In *6th International Conference on Computing, Networks and Internet of Things, CNIOT 2025, Shanghai, China, May 23-25, 2025*, pages 1–5. IEEE, 2025.

[37] Yin Yang, Keng Siau, Wen Xie, and Yan Sun. Smart health: Intelligent healthcare systems in the metaverse, artificial intelligence, and data science era. *Journal of Organizational and End User Computing (JOEUC)*, 34(1):1–14, 2022.

[38] Guangsheng Yu, Xuan Zha, Xu Wang, Wei Ni, Kan Yu, Ping Yu, J Andrew Zhang, Ren Ping Liu, and Y Jay Guo. Enabling attribute revocation for fine-grained access control in blockchain-iot systems. *IEEE Transactions on Engineering Management*, 67(4):1213–1230, 2020.

[39] Peng Zeng, Zhiting Zhang, Rongxing Lu, and Kim-Kwang Raymond Choo. Efficient policy-hiding and large universe attribute-based encryption with public traceability for internet of medical things. *IEEE Internet of Things Journal*, 8(13):10963–10972, 2021.

[40] Wei Zhang, Zhishuo Zhang, Hu Xiong, and Zhiguang Qin. Phas-hekr-cp-abe: partially policy-hidden cp-abe with highly efficient key revocation in cloud data sharing system. *Journal of Ambient Intelligence and Humanized Computing*, pages 1–15, 2022.

[41] Yichen Zhang, Jiguo Li, and Hao Yan. Constant size ciphertext distributed cp-abe scheme with privacy protection and fully hiding access structure. *IEEE Access*, 7:47982–47990, 2019.

[42] Hui Zheng, Jun Shao, and Guiyi Wei. Attribute-based encryption with outsourced decryption in blockchain. *Peer-to-Peer Networking and Applications*, 13:1643–1655, 2020.

Yanwei Zhou received the Ph.D. degree in computer software and theory from the Shaanxi Normal University, in 2018.

He is currently an Associate Professor with the School of Computer Science, Shaanxi Normal University, China. His research interests include leakage-resilient cryptography and anonymous communication.

Kui Ma received the MS degree in computer software and theory from Shaanxi Normal University in 2024. He main research interests are cryptography and information security.

Guangjin Zhang is a master degree candidate in computer science and technology at Shaanxi Normal University. He main research interests are cryptography and information security.

Guoji Song is currently a Senior Experimentalist with the Information Center, Hebei Normal University for Nationalities, China. He main research interests is information technology.

Zirui Qiao received the Ph.D. degree in computer software and theory from Shaanxi Normal University, xi'an, China, in 2024. She is currently a Lecturer with the School of Cyberspace Security, Xi'an University of Posts and Telecommunications, China. Her main research interests are cryptography and information security.

Xianxiang Liu is a master degree candidate in computer science and technology at Shaanxi Normal University. He main research interests are cryptography and information security.

Yong Yu received the Ph.D. degree in cryptography from Xidian University, Xi'an, China, in 2008.

He is currently a Professor with the School of Computer Science, Shaanxi Normal University, Xi'an, China. His research interests include information security and cryptography.